Tactix: Current-Source Sweep Core Proof Regression

Report to Andrew

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Outcome

The Sweep original-core-unit mechanism identified in the September 7 parity investigation is implemented, but lacked a maintained current-source proof fixture. AM406 adds one without changing solver semantics or the public rule registry. A four-variable saved-core row is exposed to the existing capture and emission transaction after three host root units make its original clause unit. The regression checks both sides of the eligibility boundary: with only two of those root units, it captures nothing.

Proof and test evidence

On hwulab, job `1721170` completed `0:0` in 19 seconds using two CPUs and 8 GiB. The full Sweep unit test passed, including 86,501 small environments and 5,344 checked claims. Its proof script verified 48 derivations through independent DRAT and LRAT checkers; six were also checked through the strict native DSR-to-LSR-to-Trestle route. The new original-core case is emitted in all three formats against the exact original CNF plus a checker-only contradictory unit. The build hash, exact command and job log are retained under `benchmarks/probes/am406-sweep-core-proof/`; the technical report `2026-09-25-am406-sweep-original-core-proof-regression.md` records the boundary and limitations.

Two earlier diagnostic jobs are accounted for, not hidden. The first fixture made the original unit before environment construction, so the builder correctly excluded it; the real mismatch occurs after import during a seed. The second invoked the unrelated brute-force oracle with zero rounds and hit its pre-existing nonempty-environment assertion. The final job uses the saved-core boundary and the standard round count. No new solver defect was established by either failed harness invocation.

Remaining gates

This is an adapter proof regression with a posed saved core, not a replay of the full seed-563 trajectory; the isolated historical trace remains separate. The N4 row is implemented and proof-tested but not accepted until cumulative correctness/performance gates and trajectory checks pass. The current Opus fuzz and full Debug jobs continue. H2's checker contract and the post-finalization clause-addition API decision remain open. Production online learning still requires the previously identified inprocessing controller and live strategy safe points. N1–N4 and Rank/B+C/WV gates are not complete, so the requested full Tier-D 5,000-second, four-CPU, 32-GiB comparison remains deferred.